

CMS Internal

pp $\sqrt{s} = 5.02$ TeV (28.0 pb $^{-1}$)

Events

J/ ψ data $20.0 < p_T < 25.0$, $|y| < 1.6$ $0.068 < \text{ctau3D} < 0.075$ mm

status/cov = 0/3

—•— Data

— Fit

- - - Signal

- - - Background

 $N_{\text{sig}} = 510.5 \pm 24.4$ $N_{\text{bkg}} = 164.8 \pm 15.8$

SR entries = 528

 $\mu = 3.0945 \pm 0.0028$ $\sigma_{\text{G1}} = 0.03667 \pm 0.0046$ $\sigma_{\text{CB1}} = 0.0227 \pm 0.003$

160

140

120

100

80

60

40

20

0

5

0

-5

 $\chi^2/\text{ndf} = 70.4/72$ $m_{\mu\mu}$ [GeV/c 2]

3.5

3.4

3.3

3.2

3.1

3.0

2.9

2.8

2.7

2.6